Takeoff Automaton Updated

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$$x + x_0 + \frac{1.1 v_{stall} + 1.1 v_{stall} + a\,dt}{2} dt < l_{runway}$$

$$\Rightarrow \frac{1.1 v_{stall} + v_0}{2} t_{rotate} + x_0 + \frac{2(1.1 v_{stall}) + a\,dt}{2} dt < l_{runway}$$

$$1.1 v_{stall} \leq v_0 + a\, t_{rotate}$$

$$\Rightarrow \frac{1.1 v_{stall} - v_0}{a} \leq t_{rotate}$$

$$\Rightarrow \frac{1.1 v_{stall} + v_0}{2}\left(\frac{1.1 v_{stall} - v_0}{a}\right) + x_0 + \frac{2(1.1 v_{stall}) + a\,dt}{2} dt < l_{runway}$$

Reaction time error adding

$$\frac{2(1.1 v_{stall}) + a\,dt}{2} dt$$

$$\frac{(1.1 v_{stall})^2 - v_0^2}{2a} + x_0 + \left[2(1.1 v_{stall}) + a\,dt\right] dt < l_{runway}$$

$$\int_0^{dt} (last\ v) + at \;\; d(t) = \Delta x_{dis}$$

$$\Rightarrow (last\ v)dt + \tfrac{1}{2} a\, dt^2 = \Delta x_{dis}$$

$$\int_0^{dt} v_0 + at \;\; d(t) = \Delta x_{cont}$$

from kinematics & the integral

$v_0 dt + 1/2 a dt^2 = \Delta x_{cont}$

In our model at each discrete step last $v = v_0$

$\Rightarrow \Delta x_{cont} - \Delta x_{dis} = 0$

$\Rightarrow$ No discretization error

So only reaction time needed in safety condition

$$\boxed{\frac{(1.1 v_{small})^2 - v_0^2}{2a} + x_0 + \left[2(1.1 v_{small}) + a dt\right] dt < l_{runway}}$$

Safety condition of ceiling height already enforced in guard condition.

$h < h_{ceiling}$

$\Rightarrow h_0 + c(t_{cruise} - t_{rotate}) < h_{ceiling}$ (since climb starts after t_{rotate})

$\Rightarrow$ last $h + c dt < h_{ceiling}$

$\Rightarrow$ last $h < h_{ceiling} - c dt$

Transition happens after this inequality fails:

last $h \geq h_{ceiling} - c dt$

Reaction time error addition

$c dt$

Jump condition = guard condition

$\Rightarrow$ last $h \geq h_{ceiling} - 2 c dt$

Now need to ensure velocity @ cruise exceeds $1.1 v_{stall,clean}$ (not $v_{stall,flaps}$ since $v_{stall,flaps} < v_{stall,clean}$)

$$v > 1.1 v_{stall,clean}$$

$$\Rightarrow \quad v_0 + a t_{cruise} > 1.1 v_{stall,clean}$$

$$h_0 + c\left(t_{cruise} - t_{rotate}\right) < h_{ceiling} - 2c\,dt$$

$$\Rightarrow \quad t_{cruise} < \frac{h_{ceiling} - h_0}{c} - 2c\,dt + t_{rotate}$$

$$v_0 + a\left(\frac{h_{ceil} - h_0}{c} - 2c\,dt + t_{rot}\right) > 1.1 v_{stall,clean}$$

$$\Rightarrow \quad \frac{h_{ceiling} - h_0}{c} > \frac{1.1 v_{stall,clean} - v_0}{a} + 2c\,dt - t_{rot}$$

$$\Rightarrow \quad h_{ceiling} > c\left(\frac{1.1 v_{stall,clean} - v_0}{a}\right) + 2\,dt - c\left(\frac{1.1 v_{stall,flaps} - v_0}{a}\right) + h_0$$

$$vel = last\ vel + a\,dt$$

$$last\ vel = vel - a\,dt$$

$$x = last\ x + \frac{(last\ vel + vel)}{2}\,dt$$

$$last\ x = x - \frac{(last\ vel + vel)}{2}\,dt$$